Product Information
Acronal® PLUS 7018

EC-Lock Technology for best-in-class rain-mark resistant exterior coatings

Acronal® PLUS 7018 functionalized polymer dispersion was developed with the latest EC-Lock Technology. Effectively addresses the early rain streak marks, and delivers high durability and outstanding color retention for Exterior Coatings.

Chemical Nature:
Aqueous Straight acrylic copolymer dispersion

Benefits

- EC-Lock Technology for best-in-class rain-mark resistance, especially in dark colors
- High dirt pick-up resistance
- Excellent color durability
- Excellent water & Efflorescence resistance

Features

- Excellent weatherability
- APEO free
- Excellent water resistance
- Fine particle size

Properties

Product specification*	Solids content	%	45 ± 1
	pH value	pH	8.0 – 10.5
	Viscosity at 23 °C, 2/20 rpm	mPa s	50 – 500
Other properties of dispersions	Minimum film-forming temperature (ISO 2115)	°C	24
	Density (ISO 2811-1)	g/cm^3	approx. 1.04
	Resistance to frost	ºC	≤ 0
	Type of dispersion		Anionic & non-ionic

*The aforementioned data shall constitute the agreed contractual quality of the product at the time of passing of risk. The data are controlled at regular intervals as part of our quality assurance program. Neither these data nor the properties of product specimens shall imply any legally binding guarantee of certain properties or of fitness for a specific
purpose. No liability of ours can be derived therefrom.

Note
The information submitted in this publication is based on our current knowledge and experience. In view of the many factors that may affect processing and application, these data do not relieve processors of the responsibility of carrying out their own tests and experiments; neither do they imply any legally binding assurance of certain properties or of suitability for a specific purpose. It is the responsibility of those to whom we supply our products to ensure that any proprietary rights and existing laws and legislation are observed.

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Applications

Areas of application

Acronal® PLUS 7018 is ideal for use in premium exterior coatings for best-in-class rain-mark resistance, especially in dark colors.

Processing

It is advisable to disperse the pigments and extenders with wetting and dispersing agents such as Dispex® CX 4231 and water-soluble polyphosphates in an alkaline medium in advance before the polymer dispersion is added. It is only when the products with very high viscosity are being mixed in low-speed mixers that Acronal® PLUS 7018 should be added together with the wetting and dispersing agents.

Acronal® PLUS 7018 has good pigment binding power and very good compatibility with pigments and extenders.

Various thickeners can be added to emulsion paints in order to adjust their viscosity and workability. Cellulose ethers, polyacrylates, diurethane thickeners (such as Rheovis® 1125, Rheovis® PU 1280 and Rheovis® PE 1230) and bentonite can be used. The choice of thickener depends on whether the coating is expected to be free-flowing or more thixotropic.

Solvents are not needed due to excellent film forming ability at ambient temperature. Based on specific application desires, solvent can also be added as appropriate. Short-chain alcohols and glycols improve the freeze-thaw resistance of paints but they cannot be used to lower the film-forming temperature. If possible, solvents should not be added directly to the polymer dispersion, they should be mixed in the millbase and then added at let down stage.

Like all finely divided polymer dispersions, Acronal® PLUS 7018 has a tendency to foam. It is therefore necessary to add a commercial defoamer (e.g. Foamaster® MO NDW or Foamstar® ST 2410 the level of 0.3-1.0%. Trials should be carried out to test the effectiveness of the defoamer.

Acronal® PLUS 7018 itself is resistant to microorganisms in the form in which it is supplied, but preservatives need to be added to products formulated with Acronal® PLUS 7018 to protect them from attack by microorganisms over long periods in storage. Trials should always be carried out to test the compatibility and efficacy of the preservatives.

Customers have to carry out their own trials when developing and processing products based on Acronal® PLUS 7018. The compatibility of Acronal® PLUS 7018 with other ingredients of formulations, its effect on mixing processes and its adhesion on different substrates etc., are affected in our own trials. This includes testing its stability by storing it at ca. 50° C to confirm that its viscosity remains stable.

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Safety

General
The usual precautions for handling chemicals must be observed. These include the measures set out in the guidelines of the organizations responsible for safety at work, in particular, good ventilation and fume extraction at the workplace, care of the skin and the wearing of eye protection.

Safety Data Sheet
When using this product, the information and advice given in our **Safety Data Sheet** should be observed. Due attention should also be given to the **precautions** necessary for handling chemicals.

Labelling
According to all the data at our disposal, **Acronal PLUS 7018** does not need to be labelled as a dangerous substance or preparation as defined in the relevant local directives according to their current status.

Storage

Acronal® PLUS 7018 must not be allowed to come into contact during storage with metals or alloys that are susceptible to corrosion. It is important to ensure that containers are kept tightly sealed, and the headspace of bulk storage tanks must be kept saturated with water vapor. This product must not be exposed to high temperatures, and it must be protected from frost. The product should not be exposed to direct sunlight for long periods of time.

Acronal® PLUS 7018 has a shelf life of nine months at 10 – 30 °C, provided due attention is paid to the hygiene of tanks and storage facilities.

We would recommend treating this product with a biocide in order to prevent problems with micro organisms from occurring during storage and processing.

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